

Task value inside the AI time-horizon frontier

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Abstract

METR measures the capability of AI agents by the maximum length of a task they can reliably complete. This paper estimates the share of the US wage bill comprising tasks inside that capability frontier, alongside the share representing tasks workers actively delegate to AI. By attaching a wage-bill share and a human duration estimate from the Anthropic Economic Index to each of the 18,796 O*NET work tasks, I calculate the expected success probability implied by METR’s fitted logistic curves. Under the latest model METR has measured, tasks carrying 60 per cent of the wage bill are expected to succeed. At the occupational level, no single occupation has 90 per cent of its task value inside the frontier, indicating that AI covers parts of most jobs rather than the entirety of a few. These estimates rely on two structural assumptions: equating AEI’s duration estimates for delegated chat tasks with METR’s laboratory completion times, and extrapolating success curves from software engineering to the broader economy.

1 Introduction

METR measures the performance of AI agents by the length of the tasks they can complete, expressed as the time a person would need for the same task (Kwa et al., 2025). On METR’s suite, this time horizon has grown exponentially. It doubled roughly every seven months from 2019 to early 2025, and every 4.2 months since 2023 (Kwa et al., 2025; METR, 2026). Meanwhile, the Anthropic Economic Index (AEI) tracks which work tasks people bring to Claude and estimates how long that delegated work would have taken a person without AI (Handa et al., 2025; Anthropic, 2026). This paper places both metrics on a single scale—US work tasks weighted by their corresponding wage bill—to answer two questions: What share of the US wage bill consists of tasks a frontier agent is expected to complete? And what share consists of tasks that workers actually delegate to an AI system?

Translating METR’s time horizon into an economic quantity requires weighting tasks by the wages paid for them, while mapping observed usage isolates theoretical capability from actual workplace delegation. Scenario models of AI and the labour market, such as the macroeconomic framework developed at Anthropic by Korinek et al. (2026), rely on assumed parameter values for both of these quantities to forecast economic impacts. For instance, to generate their “extreme” 2030 scenario, their model assumes an affected task mass (m_t) covering 50 per cent of all tasks, combined with an automation share (ψ_t) of 0.90, meaning nine in ten AI-performed tasks require no human supervision. By turning these theoretical dials, their framework produces massive variation in

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macroeconomic outcomes: depending on the assumptions chosen, results range from negligible labor disruptions to a scenario featuring 17.9 per cent cognitive unemployment, 15.4 per cent annual GDP growth, and the labor share of income plummeting to 45.2 per cent. Because the future trajectory of the macroeconomy hinges entirely on the values of these assumed parameters, establishing their empirical baseline is critical. This paper provides those baseline estimates.

To evaluate capability, I utilize METR’s fitted logistic curves, which map the probability of an agent’s success to the logarithm of a task’s human time. The 50 per cent horizon is the task length at which that curve crosses one half; the stricter 80 per cent horizon is the length at which it crosses four fifths. Because success falls gradually with length, the 80 per cent horizon is several times shorter than the 50 per cent horizon. Rather than relying on a single threshold point, I integrate the entire curve: for each task, I calculate the probability of success given its length and sum the wage bill weighted by that probability. I refer to this as the expected-success share, which is reported alongside the threshold shares.

I use two frontier models throughout, as METR warns that horizons above 16 hours are not reliable on its current suite. Claude Opus 4.6, released in February 2026, is the most capable model falling inside that reliability range; Mythos Preview (early), released in April 2026, is the latest measured model and sits above it. Under Claude Opus 4.6 the expected-success share is 52 per cent of the US wage bill; under Mythos Preview (early) it rises to 60 per cent. By contrast, tasks that appear in Claude usage carry just 25 per cent of the wage bill in May 2026 and 44 per cent cumulatively across all AEI releases since December 2024. Capability and observed usage do not perfectly overlap: workers frequently use Claude to augment pieces of complex tasks that the AI cannot complete autonomously, while many tasks well within the frontier agent’s capability remain entirely untouched by users. Aggregating from tasks to occupations, 64 per cent of employment sits in occupations where at least half of the task value is expected to succeed under Mythos Preview (early), yet zero per cent sits in occupations where 90 per cent succeeds.

This paper bridges two literatures. Task-based models of technical change treat a job as a bundle of tasks and ask which tasks a technology can perform (Autor et al., 2003; Acemoglu and Restrepo, 2018). Measures of occupational exposure to AI traditionally score these tasks using expert or model-based judgement (Brynjolfsson et al., 2018; Webb, 2020; Felten et al., 2021; Eloundou et al., 2023). Unlike previous literature that relies on subjective rubrics to estimate exposure, this paper grounds its estimates in empirical benchmark performance and observed user behavior. The tradeoff for this empirical grounding is that the measurements are drawn from specific, narrow domains (software evaluation and chatbot logs) and must be extrapolated to the broader economy.

Consequently, the results rely on two main structural assumptions. First, an AEI estimate of how long a delegated piece of work would take a person stands in for METR’s measure of full task length. Second, METR’s success curve, estimated predominantly on software and research tasks, is assumed to apply uniformly to other work of the same length.

Section 3 details both assumptions and provides the available evidence for the latter. Section 5.2 and Appendix F demonstrate how the aggregate estimates shift when these assumptions are varied. Section 2 describes the data, Section 3 the measures, Section 4 the results, and Section 5 discusses occupational aggregation, structural limitations, and concluding implications.

2 Data

2.1 METR time horizons

Figure 1 reproduces METR’s headline capability estimates. Each point represents a model’s 50 per cent time horizon, with filled points denoting models that established a new capability frontier at release. METR explicitly flags measurements above 16 hours as unreliable due to a scarcity of long-duration tasks in their current evaluation suite. Mythos Preview (early) is the only frontier model positioned above this 16-hour threshold, while Claude Opus 4.6 is the last model reliably measured below it; therefore, both models are utilized throughout this paper to bound the estimates.

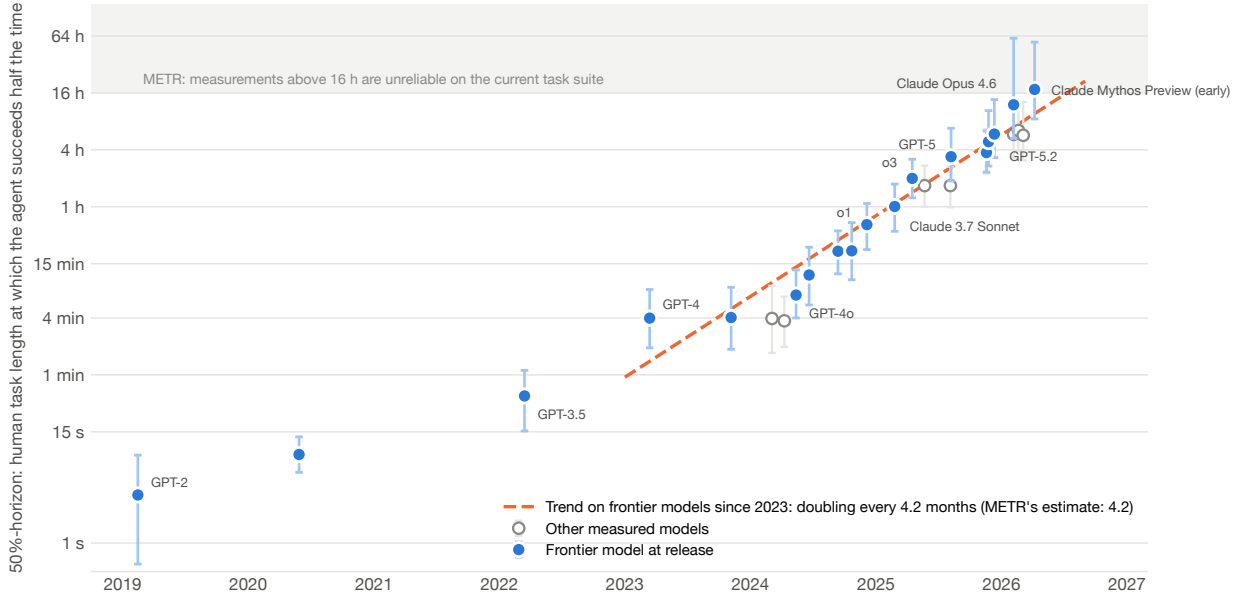


Figure 1: METR’s 50 per cent time horizons, reproduced from METR’s published estimates (METR, 2026). Bars are METR’s 95 per cent confidence intervals. Filled points: frontier model at release. Dashed line: exponential trend fitted to frontier models released from January 2023 with horizons below 16 hours. Grey band: horizons METR describes as unreliable on the current suite. GPT-2, davinci-002 and GPT-3.5 are from Time Horizon 1.0.

The evaluation suite underlying these estimates is highly specialized. METR’s published per-task results cover 228 tasks across 79 families (Appendix Table 4). All are computer-based. The majority derive from HCAST—a repository of software engineering, machine learning, cybersecurity, and research tasks with completion times benchmarked on human contractors (Rein et al., 2025). The remainder consists of research-engineering environments from RE-Bench (Wijk et al., 2024) and software actions taking mere seconds. The median task requires a human 13 minutes, and the longest 30 hours. Appendix C details the largest task families.

METR publishes each model’s fitted slope and intercept, horizons, and confidence intervals online (METR, 2026). To construct the expected-success shares, I rebuilt every run from the raw per-task success counts and refit the logistic curves using METR’s precise methodology. This refit reproduces METR’s published horizons to within 0.02 per cent for every model (Appendix Table 5), allowing me to use the refit coefficients for the full success curve while retaining METR’s published confidence intervals.

2.2 The Anthropic Economic Index

The AEI reports the share of Claude conversations mapped to specific O*NET tasks, alongside an AI-generated estimate of the human time required to complete the delegated work without AI (Handa et al., 2025; Anthropic, 2026). For the usage measure, I pool seven Claude.ai data windows spanning December 2024 to May 2026. For task durations, I rely on the June 2026 release as the baseline because it identifies tasks by their exact O*NET task IDs, allowing for a perfect merge. Earlier releases identified tasks by text matching and yielded durations shorter by a factor of approximately 1.7; these earlier vintages are retained for robustness checks (Section 5.2).

Durations are directly observed for 2,814 tasks, representing 26 per cent of the wage bill. For the remaining unobserved tasks, I impute the duration using the geometric mean of observed durations among tasks sharing an O*NET detailed work activity. If no match exists, I sequentially back off to the intermediate work activity, and finally the generalized work activity (Appendix A). The resulting wage-weighted median task duration across the economy is 3.1 hours.

2.3 Wage weights from O*NET and OEWS

To translate task durations into an economic aggregate, I map each O*NET task to its parent occupation. I assign occupation o its OEWS May 2025 total employment E_o and mean annual wage W_o (U.S. Bureau of Labor Statistics, 2026; National Center for O*NET Development, 2026). The occupational wage bill is then distributed across its core tasks in proportion to their O*NET importance ratings. Formally, task t receives a wage weight $V_t = E_o W_o s_{ot}$, with the shares s_{ot} summing to one within the occupation. Summed over all 18,796 tasks, this yields \$10.72 trillion, representing the aggregate wage bill of 153.4 million jobs. Throughout this paper, “share of the wage bill” refers to a share of this total denominator unless otherwise specified.

2.4 Tasks, work activities and the physical filter

O*NET describes occupations hierarchically: each occupation comprises a list of tasks, and each task requires a specific set of detailed work activities. Registered Nurses, for example, perform 27 tasks, ranging from recording patients’ vital signs to administering anaesthetics (Figure 7 in Appendix C). Each task maps to one or more detailed work activities, which in turn roll up into 41 generalized work activities.

I classify a generalized work activity as physical if it falls into one of seven categories: performing general physical activities, handling and moving objects, controlling machines and processes, operating vehicles and equipment, repairing mechanical equipment, repairing electronic equipment, or assisting and caring for others (see Appendix A for exact codes).

The baseline estimates employ a deliberately strict physical filter: a task is classified as physical if *any* of its detailed work activities map to a physical generalized work activity. It is classified as non-physical only if *none* do. Table 1 illustrates this mapping. Recording a patient’s medical information maps only to documenting information (non-physical). However, administering medications and monitoring patients for reactions maps to both monitoring processes (non-physical) and assisting/caring for others (physical); because it contains a physical component, the entire task is classified as physical.

This strict rule ensures that the expected-success share genuinely measures the autonomous com-

pletion of tasks exactly as O*NET defines them. Because a software agent lacks a physical embodiment, it cannot complete a mixed task unless an employer first unbundles the task and reassigns the physical component to a human worker.

Of the 18,796 tasks in the economy, 4,819 are entirely physical and 657 mix physical and non-physical activities. Together, these carry 22 per cent of the wage bill (with mixed tasks accounting for 3.3 percentage points). The remaining 13,320 tasks have no physical activity whatsoever.

I also test an alternative, lenient rule, which classifies a task as physical only if *every* work activity is physical. This assumes costless unbundling, allowing the non-physical components of mixed tasks to be instantly handed to an AI. Adopting this lenient rule raises the expected-success share under Mythos Preview (early) from 60 to 63 per cent (Appendix F). The two rules bracket the truth, and the gap between them is small relative to the other uncertainties modeled in the paper.

Physical tasks are never counted as being inside the AI capability frontier. Throughout the results, where a number is reported as a share of “all tasks,” physical tasks are included in the denominator but excluded from the numerator. Where it is a share of “non-physical tasks,” they are excluded from both. Table 2 lists the task universes used.

Table 1: From task to work activity: the O*NET mapping for the highest-importance task in each class for Registered Nurses, and for the highest-importance physical task of Construction Laborers. A task is physical if any detailed work activity it maps to sits under a physical generalized work activity.

Task (occupation)	Detailed work activities → generalized work activity	Counted as
Record patients’ medical information and vital signs. (Registered Nurses)	Record patient medical histories. → Documenting/Recording Information	non-physical
Administer medications to patients and monitor patients for reactions or side effects. (Registered Nurses)	Monitor patient conditions during treatments, procedures, or activities. → Monitoring Processes, Materials, or Surroundings Administer non-intravenous medications. → Assisting and Caring for Others (physical)	physical
Provide health care, first aid, immunizations, or assistance in convalescence or rehabilitation in locations such as schools, hospitals, or industry. (Registered Nurses)	Immunize patients. → Assisting and Caring for Others (physical) Treat acute illnesses, infections, or injuries. → Assisting and Caring for Others (physical)	physical
Tend pumps, compressors, or generators to provide power for tools, machinery, or equipment or to heat or move materials, such as asphalt. (Construction Laborers)	Operate pumps or compressors. → Controlling Machines and Processes (physical)	physical

Table 2: Task universes. Shares are of the \$10.72 trillion wage bill attached to all tasks.

Universe	Tasks	Share of wage bill
All O*NET tasks in occupations with OEWS employment and wages	18,796	100.0%
Non-physical tasks (no physical work activity)	13,320	78.3%
Tasks with an observed AEI duration (June 2026 AEI release, Apr and May 2026)	2,814	25.6%
Computer and Mathematical occupations (in-domain check, SOC 15)	770	5.9%

3 Measures

Let D_t denote task t 's duration in human hours, and $P_d(D)$ the fitted success curve of the frontier model at date d (defined as the model with the longest 50 per cent horizon released by that date). The expected-success share is

$$S_d = \frac{\sum_t V_t P_d(D_t) \mathbf{1}[t \text{ non-physical}]}{\sum_t V_t}.$$

The threshold shares, F_d^{50} and F_d^{80} , replace the continuous probability $P_d(D_t)$ with a binary indicator taking the value of one if D_t falls below the 50 or 80 per cent horizon, respectively. Similarly, the observed-use share U replaces $P_d(D_t)$ with an indicator for whether the task appears in AEI usage data, either within a specific data window or cumulatively to date.

Because S_d integrates the full probability curve, it can diverge from the strict threshold shares. When the majority of tasks exceed the 50 per cent horizon, S_d captures partial success probabilities that F^{50} zeroes out, resulting in $S_d > F^{50}$. Conversely, when most tasks are shorter than the horizon, F^{50} counts them entirely while S_d discounts them by their remaining failure probability, causing $S_d < F^{50}$.

While wage weights and AEI durations exist for the entire O*NET task universe, the success probability relies on taking METR's curve out of its original domain. METR has measured success on approximately two hundred computer tasks, yet the aggregate shares require a success probability for each of the 18,796 O*NET tasks. Applying this curve to the broader economy requires a strong structural assumption: that at a given human duration, an AI agent's probability of completing a paralegal's or a nurse's task equals the probability predicted for a software engineering task of identical length. Because METR's data cannot empirically test this assumption outside the domain of computer work, I evaluate its plausibility using three pieces of internal evidence.

First, task length is a strong predictor of success within METR's own suite. [Kwa et al. \(2025\)](#) report that the logarithm of human time explains approximately 80 per cent of the variation in success rates across their tasks. For individual frontier models, the fit is mechanically looser: the weighted R^2 of task success rates on the pooled curve is 0.24 for Mythos Preview (early) and 0.26 for Claude Opus 4.6. Much of this residual variance is an artifact of small sample sizes (few runs per task), meaning even a perfectly specified curve would hit an R^2 ceiling of roughly 0.57 on these data. Length therefore accounts for roughly 43 per cent of the explainable variation for Mythos Preview (early) (Appendix D). The remaining variance is task-specific—at a given duration, some tasks are consistently solved while others fail entirely. However, because the expected-success share aggregates over thousands of tasks, it only requires the curve to accurately capture the *average* success rate at a given length, rather than perfectly predicting individual task outcomes.

Second, the success curve exhibits some sensitivity to the composition of the evaluation suite, but the variance is small relative to the frontier's exponential trend. For instance, estimating the curve exclusively on HCAST tasks yields 50 per cent horizons of 8.1 hours for Claude Opus 4.6 and 15.4 hours for Mythos Preview (early), compared to 12.0 and 17.4 hours on the pooled suite, primarily because seconds-long software actions anchor the left tail of the pooled curve. Furthermore, iteratively dropping any single task family from the 79 families shifts Mythos Preview (early)'s horizon strictly between 14.8 and 20.9 hours (Appendix D).

Third, METR's internal robustness checks demonstrate that while qualitative task differences shift the curve's intercept, they do not alter its fundamental trajectory over time. [Kwa et al. \(2025\)](#)

find that models perform worse on tasks raters score as “messier” (lacking clean specifications or automated feedback, thereby closer to real-world work), but the trend of capability improvement across successive models remains identical for both messy and clean tasks. A similar trend emerges on external benchmarks like SWE-bench Verified. Ultimately, this evidence confirms that length strongly predicts average success within the computational domain, but it provides no empirical guarantees for a nurse.

To mitigate this fundamental uncertainty, the empirical strategy incorporates three bounds. First, as outlined in Section 2.4, all physical tasks are strictly excluded from the capability numerator. Second, every headline table isolates an in-domain check restricted to Computer and Mathematical occupations (SOC 15, 5.9 per cent of the wage bill), where the out-of-domain assumption is most likely to hold. Finally, all results are disaggregated by occupation group, allowing readers to apply their own priors and discount groups where they find the capability assumption implausible.

4 Results

Table 3 presents the baseline estimates under both frontier models, reported for all tasks and restricted to non-physical tasks. Moving from Claude Opus 4.6 to Mythos Preview (early) increases the expected-success share of the US wage bill from 52 to 60 per cent. The share of the wage bill in tasks strictly shorter than the 80 per cent horizon exhibits extreme volatility, jumping significantly as the frontier advances and crosses the median task duration. Conversely, the 50 per cent threshold share remains nearly saturated under both models, encompassing virtually the entire non-physical wage bill.

Table 3: Headline numbers, May 2026, under the two frontier models. “Non-physical” = no physical work activity. Durations from the June 2026 AEI release, imputed where unobserved. Observed-use rows do not depend on the frontier.

Measure (share of wage bill in...)	Denominator	Frontier model	
		Claude Opus 4.6 (Feb 2026; p50 12.0 h, p80 1.2 h)	Mythos Preview (early) (Apr 2026; p50 17.4 h, p80 3.1 h)
Expected success	all tasks	52%	60%
Expected success	non-physical tasks	67%	77%
Expected success, physical tasks counted as doable	all tasks	70%	79%
Shorter than 80% horizon	all tasks	9%	31%
Shorter than 80% horizon	non-physical tasks	11%	40%
Shorter than 50% horizon	all tasks	75%	78%
Shorter than 50% horizon	non-physical tasks	96%	100%
Observed in Claude usage, May 2026	all tasks	25%	25%
Observed in any AEI release, Dec 2024 to May 2026	all tasks	44%	44%
Expected success, in-domain check	SOC 15 tasks	60%	69%
Shorter than 80% horizon, in-domain check	SOC 15 tasks	3%	17%
Observed in any AEI release, in-domain check	SOC 15 tasks	76%	76%

4.1 The Task Duration Distribution

Figure 2 illustrates the mechanics behind these threshold dynamics by overlaying METR’s fitted success curves onto the US wage-bill duration distribution. The vast majority of the wage bill falls between 15 minutes and 16 hours. Because the 50 per cent horizons of both frontier models sit to the right of this distribution mass, the p_{50} threshold captures nearly all non-physical tasks, rendering it an uninformative separator.

Crucially, this near-saturation is an artifact of the duration data, not an indicator of absolute AI capability. AEI durations measure modular pieces of work that users actively delegate in a chat interface, which are inherently short by selection: only tasks carrying 3.1 per cent of the wage bill have durations exceeding the p_{50} horizon of Claude Opus 4.6. Being strictly shorter than the p_{50} horizon only implies that the fitted curve provides better-than-even odds for a task of that length. Indeed, these same non-physical tasks yield a wage-weighted expected success of 67 per cent under Claude Opus 4.6 and 77 per cent under Mythos Preview (early), rather than 100 per cent.

Because METR explicitly flags horizons beyond 16 hours as unreliable due to suite composition, the integrated expected-success share remains the most robust aggregate measure. Threshold shares are reported alongside it, with the p_{80} share proving highly sensitive to the model’s confidence interval precisely because its horizon directly intersects the densest region of the duration distribution (Appendix E).

4.2 Capability Trajectories Over Time

Figure 3 traces the expected-success and 80 per cent threshold shares as the AI capability frontier advanced from 2023 to 2026. The expected-success share grew steadily from 15 per cent under Claude 3.5 Sonnet in late 2024 to 60 per cent under Mythos Preview (early). The final step in April 2026 is the one METR flags as extending beyond its reliable measurement range. The sharp discrete jump in the p_{80} threshold share during this step is driven by three compounding factors detailed in Appendix E: the frontier is strictly defined by the p_{50} horizon, Claude Opus 4.6 possesses an unusually flat success curve, and the p_{80} horizon crossed the mode of the task duration distribution during this transition.

4.3 Disentangling Capability and Observed Usage

Figure 4(a) contrasts theoretical capability with actual workplace adoption across each AEI data window (detailed in Appendix Table 11). While observed usage in any single snapshot window has plateaued between 21 and 27 per cent of the wage bill, the cumulative union across all releases has expanded to 44 per cent as task coverage evolves.

Figure 4(b) cross-tabulates the May 2026 wage bill by capability and usage. Crucially, observed use is not a strict subset of capability. At the 80 per cent horizon, a portion of the wage bill sits beyond the capability frontier but still appears in Claude usage, indicating that workers use AI to augment pieces of complex tasks the agent cannot complete autonomously. Conversely, a much larger portion of the wage bill falls well inside the capability frontier but remains entirely untouched by users.

Therefore, the gap between the expected-success share and observed usage cannot be interpreted as a standard technology adoption lag. The aggregate usage metric includes out-of-frontier aug-

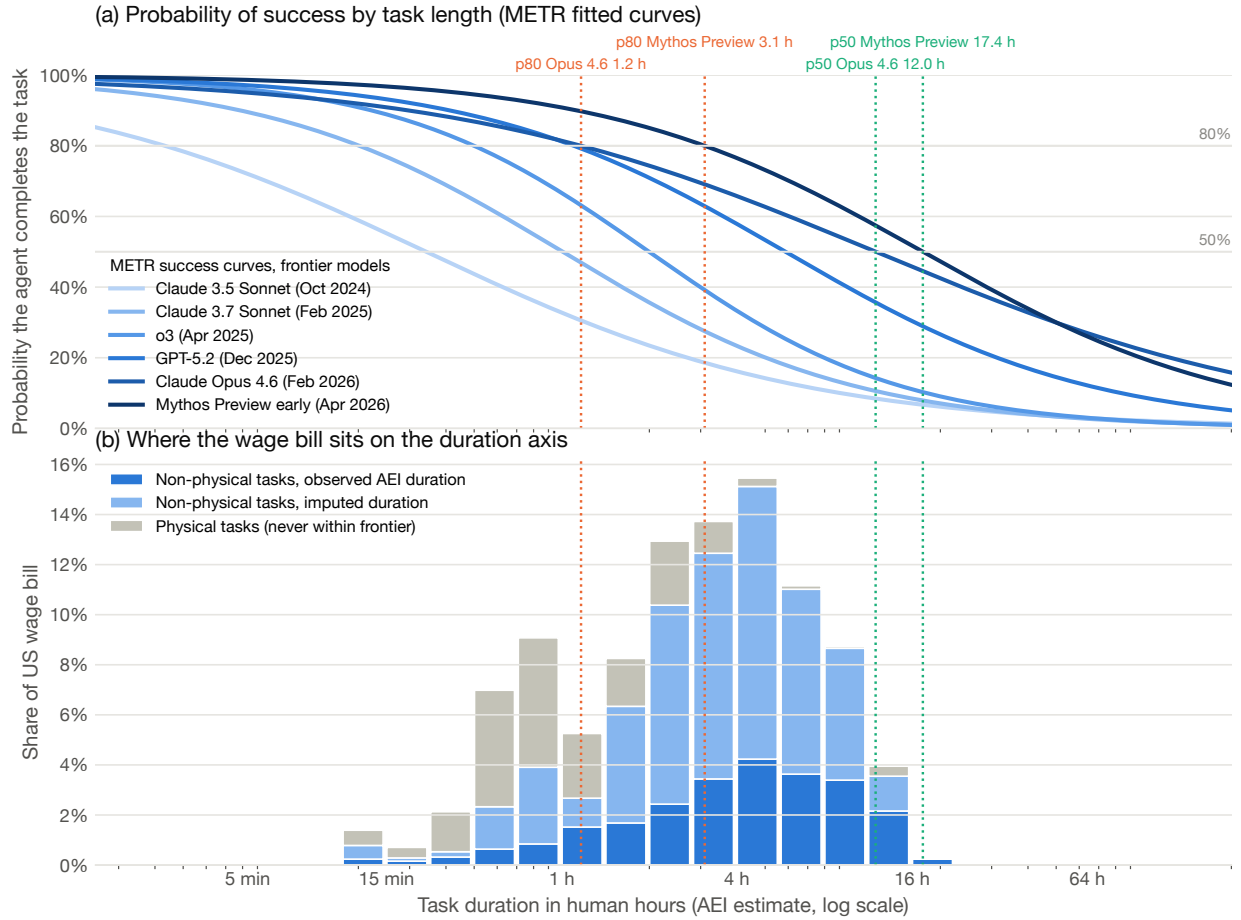


Figure 2: (a) METR’s fitted success curves for six frontier models: the probability that the agent completes a task, by the task’s human time. (b) The US wage bill by task duration on the same axis, in half-log₂ bins. Denominator: all tasks. Non-physical tasks with an observed AEI duration (dark blue), with an imputed duration (light blue) and physical tasks (grey). Dotted lines: the 50 and 80 per cent horizons of Claude Opus 4.6 and Mythos Preview (early). The two 80 per cent horizons bracket 22 per cent of the wage bill.

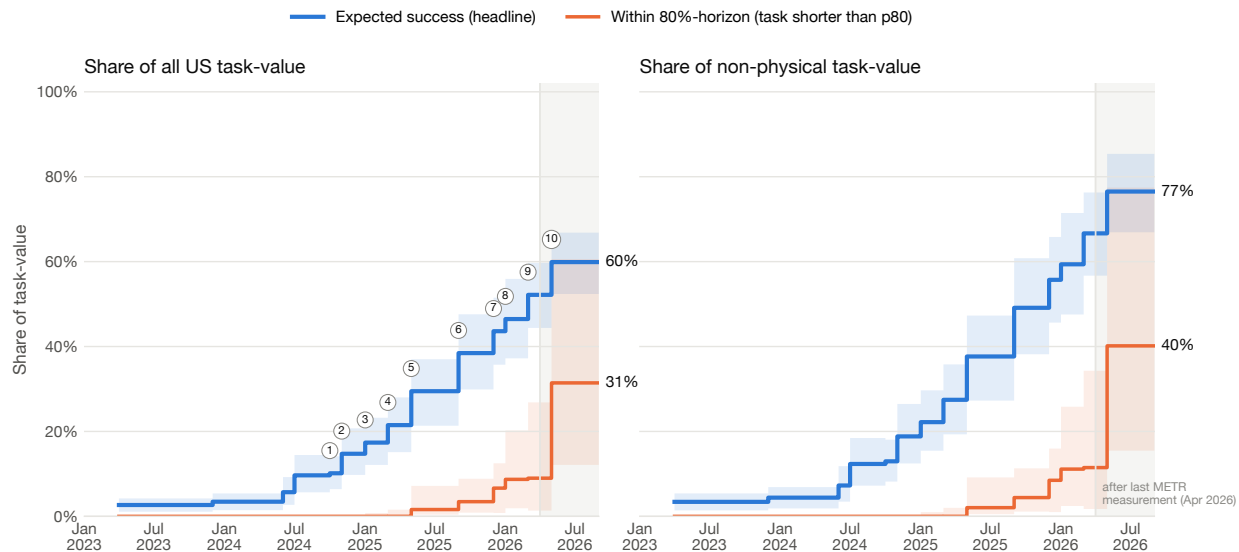


Figure 3: Share of the wage bill in tasks the frontier agent is expected to complete (blue) and in tasks shorter than the 80 per cent horizon (orange). Left: denominator is all tasks; physical tasks are in the denominator and never in the numerator. Right: denominator is non-physical tasks only. Frontier = best model released by each date, from METR’s published fits; bands = 95 per cent bootstrap intervals from the frontier model’s curve. Durations from the June 2026 AEI release, imputed for tasks without an observed duration. Grey area: after METR’s last public measurement. Numbered markers: frontier model changes, keyed in Appendix Table 5.

mentation while excluding untapped within-frontier capability. Furthermore, AEI usage data is extensive-margin only: it records whether a task appears in a chat log, not the intensity or volume of the AI’s contribution to that task.

4.4 Heterogeneity by Occupation Group

Figure 5 visualizes these three measures across the 22 SOC major groups under Mythos Preview (early). Sales, legal, office, business, and computer occupations exhibit both the highest expected-success shares and the highest observed usage. Construction, cleaning, food service, and installation occupations rank lowest, mechanically driven by the strict physical filter applied to their task value. Within the in-domain Computer and Mathematical occupations, the expected-success share reaches 69 per cent, with 76 per cent of the group’s wage bill appearing in cumulative Claude usage.

To assess the sensitivity of these baseline estimates, Appendix F varies one structural assumption at a time. Across all variants, the expected-success share under Mythos Preview (early) ranges from 49 to 75 per cent, with duration imputations driving the largest variance, while the choice of task weights within occupations barely alters the results.

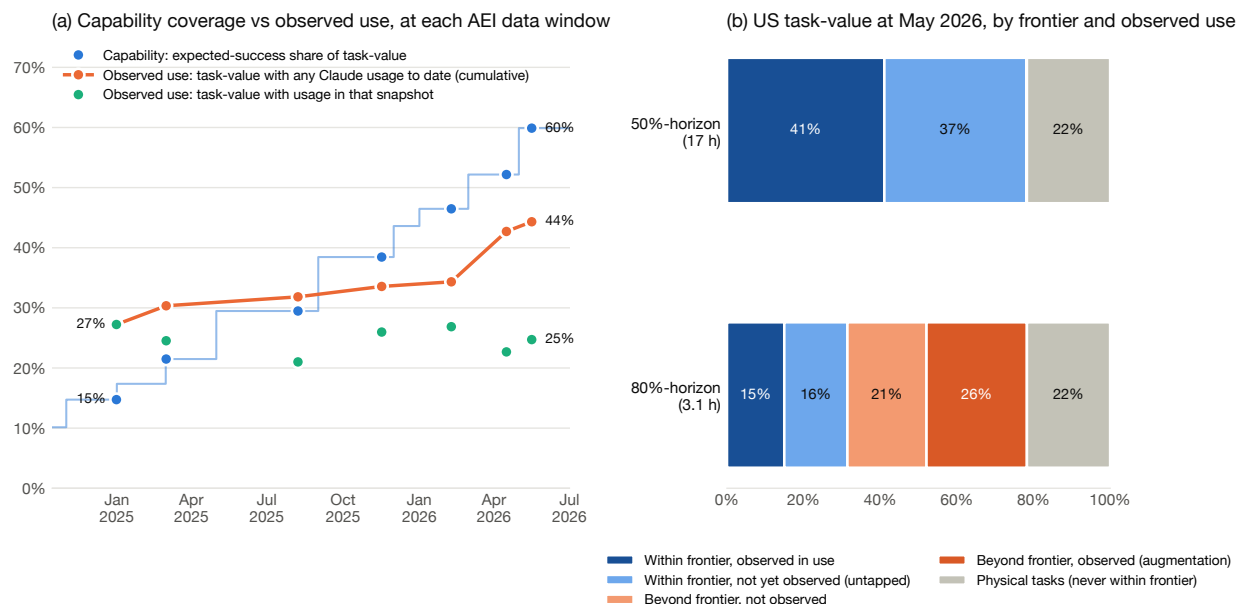


Figure 4: (a) At each AEI data window: expected-success share of the wage bill under the frontier at that date (blue); share of the wage bill in tasks with a published Claude.ai usage share in that window (green) and in any window to date (orange). Denominator: all tasks. (b) The May 2026 wage bill split by whether a task is shorter than the horizon and whether it has appeared in any AEI release, for the 50 and 80 per cent horizons of Mythos Preview (early). Physical tasks shown separately. Denominator: all tasks.

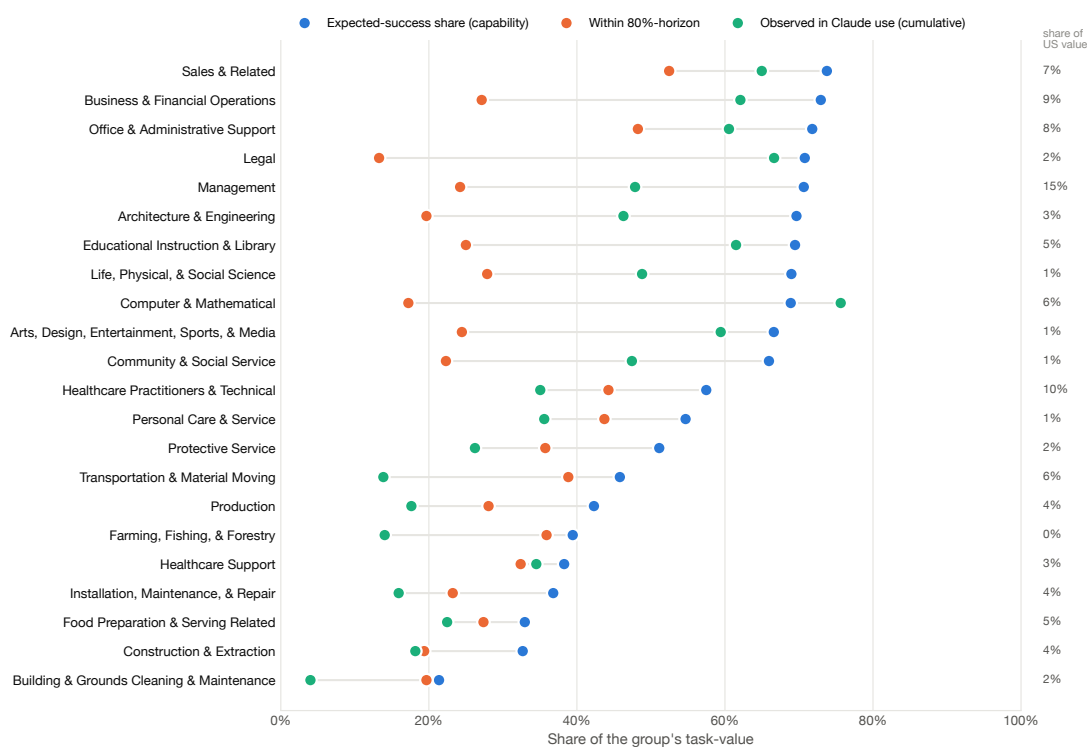


Figure 5: By SOC major group, under Mythos Preview (early): expected-success share (blue), share in tasks shorter than the 80 per cent horizon (orange), share with a published usage share in any AEI release (green). Denominator: the group's own wage bill, physical tasks included. Right column: the group's share of the total wage bill.

5 Discussion

5.1 Occupational Exposure: From Tasks to Jobs

The capability metrics calculated above represent shares of the total wage bill summed across discrete tasks. However, an occupation is a bundled set of tasks. The labor market impact of AI depends not only on the share of a job’s tasks that fall within the capability frontier, but also on whether those tasks can be cleanly unbundled. To translate task-level capability into occupational exposure, I compute each 6-digit occupation’s expected-success share and determine the proportion of US employment sitting above specific exposure thresholds.

Figure 6 illustrates this distribution for the 80 per cent threshold share across both frontier models, plotting the bootstrap band from each model’s curve (Appendix Table 10 provides the expected-success version at three thresholds). Under Mythos Preview (early), 64 per cent of US employment resides in occupations where at least half of the task value is expected to succeed, and 12 per cent where at least three quarters succeeds. Crucially, no occupation reaches the 90 per cent threshold under either model. This ceiling exists because the success curve remains below 100 per cent even at short durations, and because most occupations retain some physical, spatial, or in-person requirements. Appendix G plots every occupation’s capability against its observed use.

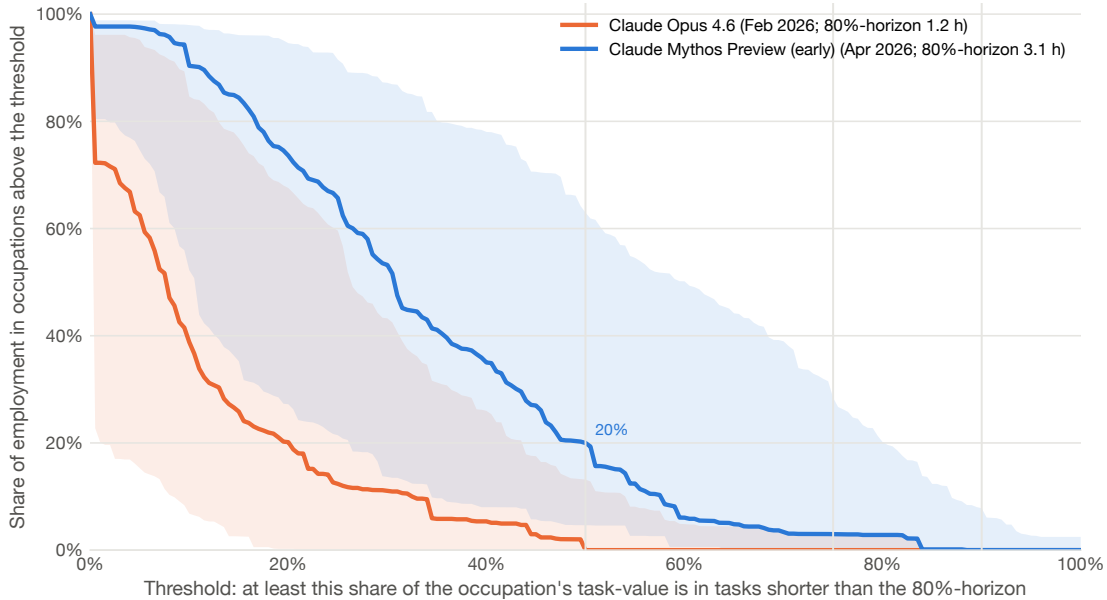


Figure 6: Share of employment in 6-digit occupations where at least the threshold share of the occupation’s task value is in tasks shorter than the frontier model’s 80 per cent horizon, for Claude Opus 4.6 (orange) and Mythos Preview (early) (blue). Denominator: 153.4 million jobs. Physical tasks count against the occupation. Bands: 95 per cent bootstrap intervals from each model’s fitted curve. Vertical lines: the thresholds in Table 10.

Therefore, an expected-success share of 60 per cent implies partial automation across most occupations, rather than the complete displacement of a few. Whether this partial coverage translates into job displacement or job augmentation depends on whether the remaining tasks serve as complements or substitutes (Autor, 2015). If the uncovered tasks constitute the core value of the job, high task-level coverage may change occupational demand very little until nearly all of it is

covered. If tasks can be cleanly peeled off and reassigned, the wage-bill share provides a closer approximation of displaced labor. Current AEI data leans toward the former reading: usage is highly concentrated in modular pieces of tasks, and those delegated pieces remain short. Resolving this definitively requires firm-level reorganization data beyond the scope of this paper.

5.2 Structural Limitations

These estimates are constrained by several structural limitations in the underlying data infrastructure:

1. **Duration Measurement Mismatch.** The two datasets run on different clocks. METR measures human time based on complete, fully specified tasks performed by contractors. AEI estimates time based on the specific, often modular pieces of work users choose to delegate in a chat interface. Delegated pieces are short by selection. If AEI durations systematically understate the length of a task as an employer would define it, the capability shares presented here are biased upward. Doubling every duration lowers the expected-success share under Mythos Preview (early) to 52 per cent and the 80 per cent threshold share to 12 per cent (Appendix F).
2. **Imputation Reliance.** Only 26 per cent of the wage bill has an empirically observed AEI duration. The imputation process carries the selection bias of delegated chat tasks into the broader occupational distribution. Restricting the analysis solely to observed tasks changes the denominator and yields an expected-success share of 73 per cent.
3. **Vintage Variance.** AEI duration estimates exhibit significant variance across releases. For the 1,451 tasks with durations in both the earlier releases and the June 2026 release, the later durations are 1.7 times longer at the median, with a log correlation of 0.67. I rely on the later release because it merges cleanly on task IDs and provides a more conservative estimate. Using the earlier releases instead raises the expected-success share by 5 points.
4. **Out-of-Domain Extrapolation.** As detailed in Section 3, METR’s success curves are applied out of domain. While task length predicts average success robustly within computational work, there is no empirical guarantee that this relationship holds across other sectors.
5. **Extensive-Margin Usage.** Observed use is Claude-only and measured on the extensive margin. AEI publishes the share of conversations mentioning a task, not the total volume or labor hours performed by AI. Furthermore, classifiers and privacy thresholds change across releases, obscuring longitudinal trends.
6. **Frontier Uncertainty.** The upper bound of the frontier is poorly measured. Mythos Preview (early)’s 50 per cent horizon of 17.4 hours carries a 95 per cent confidence interval of 8.5 to 55 hours, and METR currently treats horizons above 16 hours as unreliable. While this uncertainty has minimal impact on the expected-success share—since only 0.3 per cent of the wage bill sits above 16 hours—it introduces massive variance to the 80 per cent threshold share, whose interval runs from 12 to 61 per cent under Mythos Preview (early) (Appendix E).

By explicitly bridging METR’s laboratory success curves and AEI’s record of real-world usage onto the US wage bill, this framework isolates theoretical capability from observed workplace delegation. The resulting estimates indicate that while the capability frontier covers a majority of the wage bill in expectation, observed use covers less than half, and the overlap between the two remains highly incomplete. At the occupational level, the frontier encroaches on fractions of most jobs while fully enveloping none. The primary vulnerability in this framework remains the duration bridge separating a delegated piece of work from a fully defined employer task. Advancing this literature will require human completion times for tasks defined at the occupational level, alongside AI success

curves measured strictly outside the software domain.

References

- Daron Acemoglu and Pascual Restrepo. The race between man and machine: Implications of technology for growth, factor shares, and employment. *American Economic Review*, 108(6): 1488–1542, 2018.
- Anthropic. The Anthropic Economic Index: data releases. <https://huggingface.co/datasets/Anthropic/EconomicIndex>, 2026. Releases of February 2025 to June 2026.
- David H. Autor. Why are there still so many jobs? The history and future of workplace automation. *Journal of Economic Perspectives*, 29(3):3–30, 2015.
- David H. Autor, Frank Levy, and Richard J. Murnane. The skill content of recent technological change: An empirical exploration. *Quarterly Journal of Economics*, 118(4):1279–1333, 2003.
- Erik Brynjolfsson, Tom Mitchell, and Daniel Rock. What can machines learn, and what does it mean for occupations and the economy? *AEA Papers and Proceedings*, 108:43–47, 2018.
- Tyna Eloundou, Sam Manning, Pamela Mishkin, and Daniel Rock. GPTs are GPTs: An early look at the labor market impact potential of large language models. *arXiv preprint arXiv:2303.10130*, 2023.
- Edward Felten, Manav Raj, and Robert Seamans. Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses. *Strategic Management Journal*, 42(12):2195–2217, 2021.
- Kunal Handa, Alex Tamkin, Miles McCain, Saffron Huang, Esin Durmus, Sarah Heck, Jared Mueller, Jerry Hong, Stuart Ritchie, Tim Belonax, Kevin K. Troy, Dario Amodei, Jared Kaplan, Jack Clark, and Deep Ganguli. Which economic tasks are performed with AI? Evidence from millions of Claude conversations. *arXiv preprint arXiv:2503.04761*, 2025.
- Anton Korinek, Charles I. Jones, Szymon Sacher, Tess Cotter, and Peter McCrory. Economic scenarios for transformative AI. Working Paper 2026-02, The Anthropic Institute, 2026.
- Thomas Kwa, Ben West, Joel Becker, Amy Deng, Katharyn Garcia, Max Hasin, Sami Jawhar, Megan Kinniment, Nate Rush, Sydney Von Arx, Ryan Bloom, Thomas Broadley, Haoxing Du, Brian Goodrich, Nikola Jurkovic, Luke Harold Miles, Seraphina Nix, Tao Lin, Chris Painter, Neev Parikh, David Rein, Lucas Jun Koba Sato, Hjalmar Wijk, Daniel M. Ziegler, Elizabeth Barnes, and Lawrence Chan. Measuring AI ability to complete long software tasks. *arXiv preprint arXiv:2503.14499*, 2025.
- METR. Task-Completion Time Horizons of Frontier AI Models. <https://metr.org/time-horizons>, 2026. Time Horizon 1.1; data files `benchmark_results_1.1.yaml` and `task_results_1.1.yaml`, updated 8 May 2026.
- National Center for O*NET Development. O*NET 30.2 database. <https://www.onetcenter.org/database.html>, 2026.

David Rein, Joel Becker, Amy Deng, Seraphina Nix, Chris Canal, Daniel O’Connel, Pip Arnott, Ryan Bloom, Thomas Broadley, Katharyn Garcia, Brian Goodrich, Max Hasin, Sami Jawhar, Megan Kinniment, Thomas Kwa, Aron Lajko, Nate Rush, Lucas Jun Koba Sato, Sydney Von Arx, Ben West, Lawrence Chan, and Elizabeth Barnes. HCAST: Human-calibrated autonomy software tasks. *arXiv preprint arXiv:2503.17354*, 2025.

U.S. Bureau of Labor Statistics. Occupational employment and wage statistics, May 2025 national estimates. <https://www.bls.gov/oes/>, 2026.

Michael Webb. The impact of artificial intelligence on the labor market. Working paper, Stanford University, 2020.

Hjalmar Wijk, Tao Lin, Joel Becker, Sami Jawhar, Neev Parikh, Thomas Broadley, Lawrence Chan, Michael Chen, Josh Clymer, Jai Dhyani, Elena Elicheva, Katharyn Garcia, Brian Goodrich, Nikola Jurkovic, Holden Karnofsky, Megan Kinniment, Aron Lajko, Seraphina Nix, Lucas Sato, William Saunders, Maksym Taran, Ben West, and Elizabeth Barnes. RE-Bench: Evaluating frontier AI R&D capabilities of language model agents against human experts. *arXiv preprint arXiv:2411.15114*, 2024.

A Data and Pipeline Construction

Everything in this paper is produced by one pipeline from the raw downloads (`run_all.sh` in the `projects/value_horizon` directory of the job-transition repository); every number in the text is written by the pipeline into a macro file, and every numbered assumption is listed in `ASSUMPTIONS.md`.

- **METR:** metr.org/time-horizons, files `benchmark_results_1_1.yaml` and `task_results_1_1.yaml` (updated 8 May 2026), plus github.com/METR/eval-analysis-public for the method (`reports/time-horizon-1-1`). Refit: weighted logistic regression of run success on $\log_2$ human minutes, weights = family weight / runs per task normalised to one per model, regularisation 10^{-5} ; 500 bootstrap draws over family, task and run. Frontier = highest 50 per cent horizon among models released by the date.
- **AEI:** huggingface.co/datasets/Anthropic/EconomicIndex. Usage shares by O*NET task from seven Claude.ai data windows (Dec 2024–Jan 2025, Feb–Mar 2025, Aug 2025, Nov 2025, Feb 2026, Apr 2026, May 2026). Per-task human time from the June 2026 release (April and May 2026, simple mean of the two monthly means; baseline) and from the Nov 2025 and Feb 2026 releases (pooled; alternative). Releases before June 2026 identify tasks by text; I match text to O*NET task IDs exactly, recovering 87 to 91 per cent of usage, and drop the rest (mostly tasks of occupations since retired from O*NET). The June 2026 release carries task IDs and joins fully. Imputation: geometric mean of observed durations among tasks sharing a detailed work activity (36 per cent of the wage bill), an intermediate work activity (22 per cent), a generalized work activity (15 per cent), or all tasks (0.8 per cent).
- **O*NET 30.2:** task statements, task ratings (importance, relevance), tasks to detailed work activities, work-activity hierarchy. Weights use core tasks with positive relevance; specialties are averaged into the 6-digit SOC code; 29 occupations without ratings get equal weights. Physical work activities: 4.A.3.a.1–4 (performing general physical activities, handling and moving objects, controlling machines and processes, operating vehicles and equipment), 4.A.3.b.4–5 (repairing and maintaining equipment) and 4.A.4.a.5 (assisting and caring for others).

- **OEWS May 2025, national:** employment and mean annual wage by detailed occupation; 12 OEWS hybrid codes mapped explicitly. Occupations without O*NET tasks hold 1.3 per cent of employment and are dropped.
- **Eloundou et al. (2023) labels:** github.com/openai/GPTs-are-GPTs, used only for the alternative definition of the non-physical set.

B METR Task Suite and Refit Verification

Table 4 details the composition of METR’s Time Horizon 1.1 task suite across its three primary sources. Table 5 reports METR’s published 50 and 80 per cent horizons alongside the refitted values obtained from the per-task results across all models with fitted curves. “Frontier at release” corresponds to METR’s `is_sota` indicator; these models define the historical frontier steps depicted in Figure 3.

Table 4: Composition of METR’s Time Horizon 1.1 task suite. Weight = share of METR’s diversity-adjusted weight in the Mythos Preview (early) fit.

Source	Content	Tasks	Families	Human time: median, max	Weight
HCAST	software engineering, machine learning, cybersecurity, and computer-based research tasks	157	69	44 min, 30.0 h	81%
RE-Bench	machine-learning research engineering tasks	5	5	8.0 h, 8.2 h	4%
SWAA	seconds-long software actions (code completion, file selection, request routing)	66	5	5 s, 43 s	15%

Table 5: METR’s published 50 and 80 per cent horizons and my refit from the per-task results, all models with fitted curves. “Frontier at release” is METR’s `is_sota` flag; these models are the numbered steps in Figure 3, in date order from GPT-4 (March 2023).

Model	Released	p50 METR (min)	p50 refit (min)	p80 METR (min)	p80 refit (min)	Frontier at release
GPT-4 0314	2023-03-14	4.0	4.0	0.9	0.9	yes
GPT-4 1106	2023-11-06	4.0	4.0	0.8	0.8	yes
Claude 3 Opus	2024-03-04	4.0	4.0	0.6	0.6	
GPT-4 Turbo	2024-04-09	3.7	3.7	0.9	0.9	
GPT-4o	2024-05-13	7.0	7.0	1.3	1.3	yes
Claude 3.5 Sonnet (Old)	2024-06-20	11.4	11.4	1.7	1.7	yes
o1-preview	2024-09-12	20.3	20.3	4.4	4.4	yes
Claude 3.5 Sonnet (New)	2024-10-22	20.5	20.5	2.6	2.6	yes
o1	2024-12-05	38.8	38.8	7.1	7.1	yes
Claude 3.7 Sonnet	2025-02-24	60.4	60.4	12.1	12.1	yes
o3	2025-04-16	119.7	119.7	30.0	30.0	yes
Claude 4 Opus	2025-05-22	100.4	100.4	20.4	20.4	
Claude 4.1 Opus	2025-08-05	100.5	100.5	23.5	23.5	
GPT-5	2025-08-07	203.0	203.0	38.3	38.3	yes
Gemini 3 Pro	2025-11-18	224.3	224.3	54.1	54.1	yes
GPT-5.1-Codex-Max	2025-11-19	223.7	223.7	50.6	50.6	
Claude Opus 4.5	2025-11-24	293.0	293.0	49.4	49.3	yes
GPT-5.2	2025-12-11	352.2	352.3	66.0	66.0	yes
GPT-5.3-Codex	2026-02-05	349.5	349.5	54.7	54.7	
Claude Opus 4.6	2026-02-05	718.8	718.8	69.9	69.9	yes
Gemini 3.1 Pro	2026-02-19	384.1	384.1	89.8	89.8	
GPT-5.4	2026-03-05	341.7	341.7	53.9	53.9	
Claude Mythos Preview (early)	2026-04-07	1044.8	1044.7	185.9	185.9	yes

C Task Examples and Classification

Table 6 lists the twelve METR task families with the largest weight in the fit; the family names are METR’s identifiers, and Rein et al. (2025) and Wijk et al. (2024) describe the tasks. Table 7 shows, for eight SOC major groups, the O*NET task with the largest wage bill in the group, with its duration on the baseline axis. Figure 7 shows every O*NET task of one occupation, Registered Nurses, with the physical filter applied.

Table 6: The twelve METR task families with the largest diversity-adjusted weight in the Mythos Preview (early) fit.

Task family	Source	Tasks	Human time, shortest to longest	Weight
arithmetic	SWAA	20	1 s to 43 s	3.7%
file_selection	SWAA	20	2 s to 6 s	3.7%
alert_triage	SWAA	10	4 s to 10 s	2.6%
env_scientist	HCAST	9	44 min to 11.1 h	2.5%
request_routing	SWAA	9	8 s to 18 s	2.5%
pico.ctf	HCAST	7	2 min to 1.1 h	2.2%
data_cleaning_arjun	HCAST	7	6 min to 6 min	2.2%
code_completion	SWAA	7	3 s to 6 s	2.2%
palisade_crackme	HCAST	6	21 min to 8.0 h	2.0%
blackbox	HCAST	6	10 min to 1.1 h	2.0%
password_check	HCAST	6	57 s to 32 min	2.0%
hypothesis_testing	HCAST	6	9 min to 49 min	2.0%



Figure 7: The 27 O*NET tasks of Registered Nurses (O*NET-SOC 29-1141.00; nursing specialties excluded here). Each tile is one task statement, shortened. Dark blue: no physical work activity. Grey with orange border: some physical work activities, so counted as physical. Grey: every work activity physical. Physical tasks never count as within the frontier. Grey text: the task's AEI human duration on the baseline axis; * = imputed. Tiles are ordered by classification and then by O*NET importance.

Table 7: The O*NET task with the largest wage bill in each of eight SOC major groups. Hours = AEI human duration on the baseline axis; * = imputed. Physical = at least one of the task’s work activities is physical, so the task never counts as within the frontier.

Occupation	Task (O*NET statement)	Hours	Physical
General and Operations Managers	Review financial statements, sales or activity reports, or other performance data to measure productivity or goal. . .	7.0	
Software Developers	Analyze user needs and software requirements to determine feasibility of design within time and cost constraints.	11.3	
Lawyers	Interpret laws, rulings and regulations for individuals and businesses.	4.3	
Registered Nurses	Record patients’ medical information and vital signs.	2.0*	
Retail Salespersons	Greet customers and ascertain what each customer wants or needs.	0.5	yes
Customer Service Representatives	Confer with customers by telephone or in person to provide information about products or services, take or enter orders. . .	0.8	
Construction Laborers	Tend pumps, compressors, or generators to provide power for tools, machinery, or equipment or to heat or move materials. . .	2.1*	yes
Laborers and Freight, Stock, and Material Movers, Hand	Maintain equipment storage areas to ensure that inventory is protected.	0.6*	

D Variation within METR’s Suite

Figure 8 plots, for the two frontier models, each METR task’s observed success rate against its human time, with METR’s pooled curve and curves fitted within each task source. Table 8 gives the horizons from each fit and the range of horizons obtained by dropping any one task family and refitting. The R^2 in the panel titles is the weighted R^2 of task success rates on the pooled curve; the corresponding ceiling if the curve were exactly right, given the number of runs per task, is 0.61 for Claude Opus 4.6 and 0.57 for Mythos Preview (early). Across all 23 models with fitted curves the highest weighted R^2 is 0.68 (GPT-4 0314, ceiling 0.89); the frontier models sit low because most of their tasks are solved nearly every time and the remaining failures are concentrated in particular tasks rather than at particular lengths. The RE-Bench fits are not shown because five tasks do not identify a slope.

Table 8: Horizons from METR’s pooled fit, from fits within one task source, and the range when any one of the 79 task families is dropped. Success = diversity-weighted share of runs that succeeded.

Model	Fit	Tasks	Weight	Success	50% horizon	80% horizon
Claude Opus 4.6	pooled (METR’s fit)	228	100%	79%	12.0 h	1.2 h
	HCAST tasks only	157	81%	74%	8.1 h	1.1 h
	SWAA tasks only	66	15%	98%	1 min	36 s
	RE-Bench tasks only	5	4%	97%	n/a	n/a
	drop any one family				10.3 h to 14.2 h	1.1 h to 1.4 h
Mythos Preview (early)	pooled (METR’s fit)	228	100%	85%	17.4 h	3.1 h
	HCAST tasks only	157	81%	83%	15.4 h	3.1 h
	SWAA tasks only	66	15%	99%	16 min	4 min
	RE-Bench tasks only	5	4%	74%	n/a	n/a
	drop any one family				14.8 h to 20.9 h	2.9 h to 3.6 h

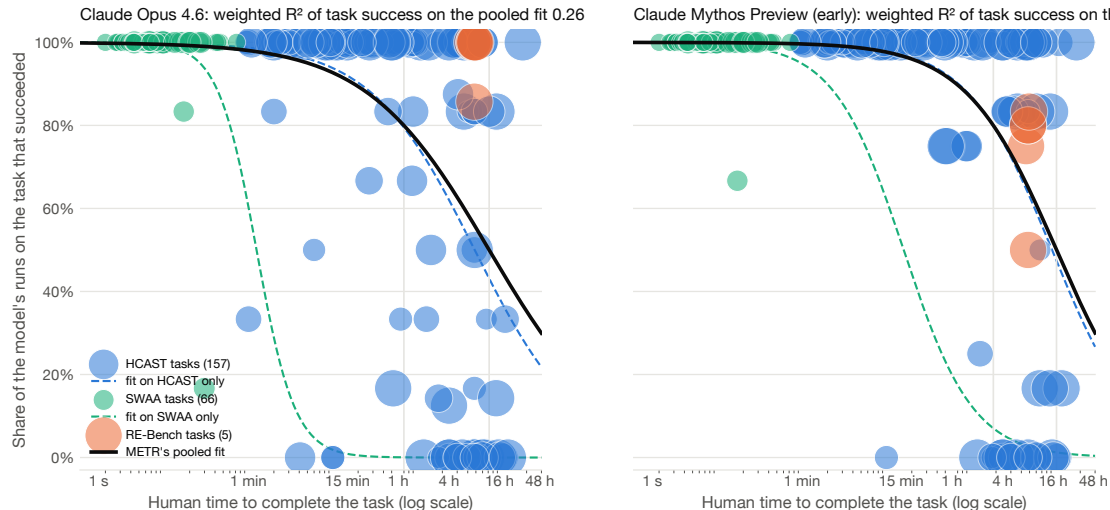


Figure 8: Each point is one METR task; area is proportional to METR’s diversity weight. Black: METR’s pooled fit. Dashed: logistic fits within one task source. Grey lines: the pooled 50 and 80 per cent horizons. Left: Claude Opus 4.6. Right: Mythos Preview (early).

E Decomposing the April 2026 Capability Step

The 80 per cent threshold share in Figure 3 jumps from 9 to 31 per cent when the frontier passes from Claude Opus 4.6 to Mythos Preview (early). The step is a faithful reading of METR’s published fits, but three things make it as large as it is. First, the frontier only moves when a model beats the best 50 per cent horizon to date. METR measured two other models between these two: Gemini 3.1 Pro (19 February 2026; 50 per cent horizon 6.4 hours, 80 per cent horizon 1.5 hours) and GPT-5.4 (5 March 2026; 50 per cent horizon 5.7 hours, 80 per cent horizon 0.9 hours). Neither beat the 50 per cent horizon of Claude Opus 4.6, so neither appears in the figure, although the 80 per cent horizon of Gemini 3.1 Pro was the longer of the two and would have put 11 per cent of the wage bill below it. Second, Claude Opus 4.6 has an unusually flat success curve. Its 50 per cent horizon is 10 times its 80 per cent horizon, against 4 to 6 times for the other frontier models since 2025, so its 80 per cent horizon barely moved from that of GPT-5.2 (1.1 to 1.2 hours) while its 50 per cent horizon doubled. Mythos Preview (early) brings the ratio back to 6. Third, the 80 per cent threshold share counts tasks shorter than a threshold, and the threshold moved from 1.2 to 3.1 hours, across the part of the duration axis where the wage bill is: 22 per cent of the wage bill is in non-physical tasks with durations between those two values (Figure 2(b)), and 32 per cent is in non-physical tasks between one and four hours. The expected-success share integrates the curve instead of counting a threshold, and moves 8 points on the same step.

The threshold is also poorly determined at the top. The 80 per cent horizon of Mythos Preview (early) has a 95 per cent interval of 1.6 to 6.6 hours, which puts the share anywhere between 12 and 61 per cent (the orange band in Figure 3). Under Claude Opus 4.6 the interval is 1 to 27 per cent. The level of the orange line in 2026 is therefore uncertain by tens of points either way, and the size of the step says as much about where the wage bill sits on the duration axis as about the model.

F Robustness and Sensitivity Analysis

Table 9 changes one assumption at a time, with Mythos Preview (early) as the frontier. The expected-success share ranges from 49 to 75 per cent. The duration bridge matters most. Doubling every duration gives 52 per cent and cuts the 80 per cent threshold share to 12 per cent; halving them raises that share to 58 per cent. Using the earlier AEI releases for durations raises the expected-success share by 5 points. Counting a task as physical only if every one of its work activities is physical, so that mixed tasks count as non-physical, gives 63 per cent. Replacing the physical-work rule with the exposure labels of Eloundou et al. (2023) gives 49 per cent. The choice of task weights within occupations barely matters, and neither does the usage threshold or the addition of API traffic to Claude.ai traffic.

Table 9: One assumption changed at a time; frontier = Mythos Preview (early). Denominator: all tasks in the universe named, which is all tasks except in the observed-duration rows.

Variant (one change from baseline)	Expected success	< p80	Observed, cumulative
Baseline	60%	31%	44%
Durations from the Nov 2025 and Feb 2026 AEI releases	64%	51%	44%
Observed-duration tasks only (denominator changes)	73%	34%	100%
Observed-duration tasks only, Nov 2025 and Feb 2026 AEI releases	75%	56%	100%
All durations halved	66%	58%	44%
All durations doubled	52%	12%	44%
Equal task weights within occupation	60%	31%	44%
Importance $\times$ relevance weights	60%	32%	45%
Frequency-implied weights	60%	32%	46%
Employment weights, no wages	55%	33%	39%
Non-physical set = Eloundou et al. exposure label > 0	49%	24%	44%
Physical = every work activity physical (lenient filter)	63%	34%	44%
Observed only if share $\geq 0.01\%$	60%	31%	20%
Observed in Claude.ai or API	60%	31%	49%

G Occupation-Level Exposure and Time Series

Figure 9 plots every 6-digit occupation by its expected-success share of task value under Mythos Preview (early) and the share of its task value that has appeared in Claude usage. Occupations fall into two clusters: office, sales, business and computer work at the top right; service, construction, production and transport work at the bottom left. Large occupations with high capability but use well below the economy-wide share, such as general managers and registered nurses, make up much of the “within the frontier, not yet observed” cell of Figure 4(b). Table 10 gives the employment shares behind Section 5.1, and Table 11 the capability and use measures at each AEI data window.

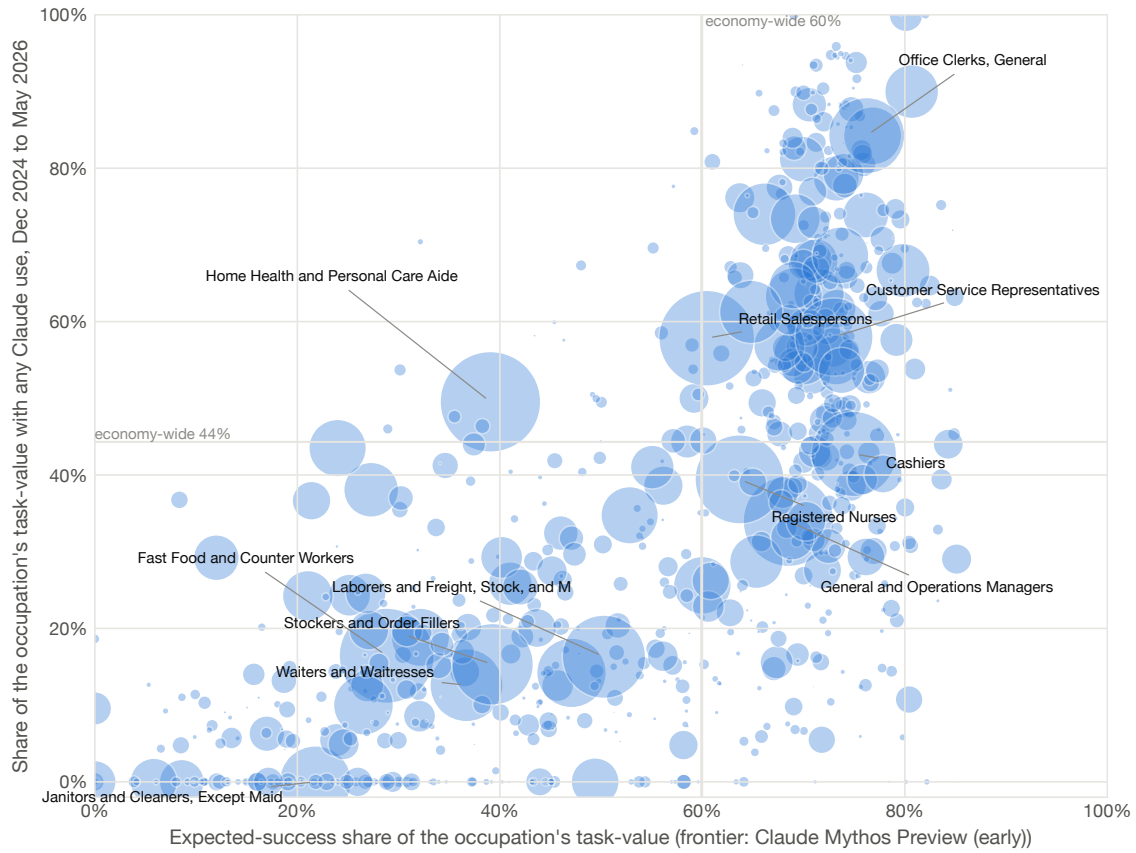


Figure 9: One bubble per 6-digit occupation (782 occupations), area proportional to employment. Horizontal axis: expected-success share of the occupation's task value under Mythos Preview (early). Vertical axis: share of the occupation's task value with a published Claude.ai usage share in any AEI release, December 2024 to May 2026. Grey lines: economy-wide values. Physical tasks count against the occupation on both axes. The 12 largest occupations are labelled.

Table 10: From tasks to jobs. Share of employment (153.4 million jobs) in occupations whose own coverage measure is at least the threshold. Physical tasks count against the occupation. Observed use is not conditioned on.

Occupation-level measure	At least	Claude Opus 4.6	Mythos Preview (early)
Expected-success share of the occupation's task value	50%	58%	64%
	75%	0%	12%
	90%	0%	0%
Share of the occupation's task value shorter than the 80% horizon	50%	0%	20%
	75%	0%	3%
	90%	0%	0%
Share of the occupation's task value shorter than the 50% horizon	50%	70%	70%
	75%	53%	54%
	90%	30%	36%

Table 11: Capability and observed use at each AEI data window. Denominator: all tasks. “< p80” = share of the wage bill in non-physical tasks shorter than the frontier model’s 80 per cent horizon.

AEI data window	Frontier model	p50 (h)	Expected success	< p80	Observed, this window	Observed, cumulative
Dec 2024–Jan 2025	Claude 3.5 Sonnet	0.3	15%	0%	27%	27%
Feb–Mar 2025	Claude 3.7 Sonnet	1.0	21%	0%	25%	30%
Aug 2025	o3	2.0	29%	2%	21%	32%
Nov 2025	GPT-5	3.4	38%	3%	26%	34%
Feb 2026	GPT-5.2	5.9	46%	9%	27%	34%
Apr 2026	Claude Opus 4.6	12.0	52%	9%	23%	43%
May 2026	Mythos Preview (early)	17.4	60%	31%	25%	44%